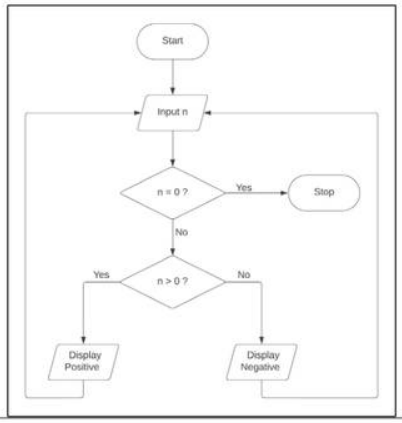
Type MS · Spot selection and iteration, and pick the right loop

Answer 1 of 4 · Question 6 part b.iv · 2 marks

2025 JUN · Paper 01 · extract 1 of 4 · find part b.iv in this mark scheme extract

Question	Answer	Additional Guidance	mark
6(a)	Award **one** mark for each of: • Start and Input n identified (accept number or num) (1) • Decision box test for n = 0/n<>0 with yes & no/true & false labels that correctly lead to End/Stop (1) • Decision box test for n > 0 or n<0 with yes & no/true & false labels (1) • Correct messages output including leads from correct yes & no/true & false (1) • Flowchart is fully connected and functionally correct (1) • Correct flowchart symbols used throughout for terminators, input/output and decision boxes (1) Example:  	Accept == for MP2 For MP2 and MP3 decision boxes must have yes/no or true/false labels. For MP5 there must be arrows on connections	6

Type MS · Spot selection and iteration, and pick the right loop

Answer 1 of 4 · Question 6 part b.iv · 2 marks

2025 JUN · Paper 01 · extract 2 of 4 · find part b.iv in this mark scheme extract

<table>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>6(b)(i)</td><td>Award one mark for each pair of columns:
<table><tr><th>number</th><th>flag</th><th>testNum</th><th>DISPLAY</th></tr><tr><td>7</td><td>FALSE</td><td>1</td><td></td></tr><tr><td></td><td></td><td>2</td><td></td></tr><tr><td></td><td></td><td>3</td><td></td></tr><tr><td></td><td></td><td>4</td><td></td></tr><tr><td></td><td></td><td>5</td><td></td></tr><tr><td></td><td></td><td>6</td><td></td></tr><tr><td></td><td>TRUE</td><td>7</td><td></td></tr></table></td><td>Accept for first pair
<table><tr><th>number</th><th>flag</th></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>FALSE</td></tr><tr><td>7</td><td>TRUE</td></tr></table>Do not award either mark point if first row is 7 FALSE 2</td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>6(b)(ii)</td><td>Award one mark for each pair of columns:
<table><tr><th>number</th><th>flag</th><th>testNum</th><th>DISPLAY</th></tr><tr><td>9</td><td>FALSE</td><td>1</td><td></td></tr><tr><td></td><td></td><td>2</td><td></td></tr><tr><td></td><td>TRUE</td><td>3</td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr></table></td><td>Accept for first pair
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</table>

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Spot selection and iteration, and pick the right loop

Answer 1 of 4 · Question 6 part b.iv · 2 marks

2025 JUN · Paper 01 · extract 3 of 4 · find part b.iv in this mark scheme extract

Question	Answer	Additional Guidance	Mark
6(b)(iii)	Award **one** mark for: • To determine if a number is a prime number/none of the test numbers divide exactly into it (1)	Ignore reference to display the number	1
Question	Answer	Additional Guidance	Mark
6(b)(iv)	Award up to **two** marks for a linked explanation such as: • It makes the algorithm more efficient (1) because the loop will terminate as soon as a test number divides exactly into the number entered (1) • A count-controlled loop could make more comparisons/be less efficient (1) because it would not terminate as soon as the test number divides exactly into the number entered (1) • A count-controlled loop executes a fixed number of times/even after condition is met (1) post-condition loop stops when condition is met (1)	Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
		Total for question 6	**13**

Type MS · Spot selection and iteration, and pick the right loop

Answer 1 of 4 · Question 6 part b.iv · 2 marks

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Type MS · Spot selection and iteration, and pick the right loop

Answer 2 of 4 · Question 1 part c · 1 mark

2021 NOV · Paper 02 · extract 1 of 9 · find part c in this mark scheme extract

1 (a)	A1	**Award 1 mark for any of:** • Comments (1) • Indentation (1) • Meaningful variable/constant/subprogram names (1) • White space (1)		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot selection and iteration, and pick the right loop

Answer 2 of 4 · Question 1 part c · 1 mark

2021 NOV · Paper 02 · extract 2 of 9 · find part c in this mark scheme extract

1 (b)	B1 B2 B3	**Award** 1 mark for each column: **Error description** / **Logic** / **Syntax** / **Runtime** Divide by 0 / / / ✓ Use x instead of * multiply / / ✓ / Subtract 10 from 2 instead of 2 from 10 / ✓ / /		(3)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot selection and iteration, and pick the right loop

Answer 2 of 4 · Question 1 part c · 1 mark

2021 NOV · Paper 02 · extract 3 of 9 · find part c in this mark scheme extract

1 (c)	C1	**The only correct answer is D** *A is not correct because as it is a data structure* *B is not correct because as there is no arrow looping back to the condition* *C is not correct because as the flowchart does not include an operator*		(1)

Type MS · Spot selection and iteration, and pick the right loop

Answer 2 of 4 · Question 1 part c · 1 mark

2021 NOV · Paper 02 · extract 4 of 9 · find part c in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot selection and iteration, and pick the right loop

Answer 2 of 4 · Question 1 part c · 1 mark

2021 NOV · Paper 02 · extract 5 of 9 · find part c in this mark scheme extract

1 (d)	**Award 1 mark for each of:** D1 = changed to == (1) D2 Condition changed to less than / final two print statements swapped (1) D3 else changed to else (1)		(3)

Code examples

C#

```
if (number1 == number2)
{
    Console.WriteLine("The numbers are equal");
}
else if (number1 < number2)
{
    Console.WriteLine("The highest number is " + number2 + " and the lowest number is " + number1);
}
else
{
```

Java

```
if (number1 == number2)
{
    System.out.print("Numbers are equal");
}
else if (number1 < number2)
{
    System.out.print("The highest number is " + number2 + " and the lowest number is " +number1);
}
else
```

Python

```
if number1 == number2:
    print("Numbers are equal")
elif number1 < number2:
    print("The highest number is", number2, "and the lowest number is", number1)
else:
```

Type MS · Spot selection and iteration, and pick the right loop

Answer 2 of 4 · Question 1 part c · 1 mark

2021 NOV · Paper 02 · extract 6 of 9 · find part c in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Spot selection and iteration, and pick the right loop

Answer 2 of 4 · Question 1 part c · 1 mark

2021 NOV · Paper 02 · extract 7 of 9 · find part c in this mark scheme extract

1 (e)	**Award 3 marks for:**			
	E1	Prompt for a number to be input (1)		
	E2	Input stored in an integer variable (1)		
	E3	Loop used (1)		
	E4	Loop iterates only for the values 1 – 12 (1)		
	E5	Display is correct when program executes (1)		(5)

Code examples

C#

```
// Print prompt and get number from user
Console.Write("Enter the number: ");
int num = Convert.ToInt32(Console.ReadLine());

// Create loop to display the table
for (int count = 1; count <= 12; count++)
{
    Console.WriteLine(count + " x " + num + " is " + count * num);
}
Console.ReadLine();
```

Java

```
// Print prompt and get number from user
Scanner input = new Scanner(System.in);
System.out.print("Enter the number ");
int num = input.nextInt();

// Create loop to display the table
for (int count = 1; count <=12; count++)
{
    System.out.println(count + " x " + num + " is " + count * num);
}
```

Type MS · Spot selection and iteration, and pick the right loop

Answer 2 of 4 · Question 1 part c · 1 mark

2021 NOV · Paper 02 · extract 8 of 9 · find part c in this mark scheme extract

Python	`# Print prompt and get number from user`
	`num = int(input("Enter the number: "))`
	
	`# Create loop to display the table`
	`for count in range(1, 13):`
	`    print(count, "x" , num , "is" , count * num)`

Type MS · Spot selection and iteration, and pick the right loop

Answer 2 of 4 · Question 1 part c · 1 mark

2021 NOV · Paper 02 · extract 9 of 9 · find part c in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Spot selection and iteration, and pick the right loop

Answer 3 of 4 · Question 6 part a · 1 mark

2024 JUN · Paper 01 · extract 1 of 5 · find part a in this mark scheme extract

6(a)	**The only correct answer is D** *A is not correct because it is not a REPEAT UNTIL / DO WHILE loop* *B is not correct because it is not a FOR loop* *C is not correct because there is no such loop*		1

6(b)(i)	Award **one** mark for any of the following:
	
	• `WHILE NOT (index >= LENGTH(arrival))`
	• `WHILE NOT (index > LENGTH(arrival)-1)`
	• `WHILE index < LENGTH(arrival)`
	• `WHILE index <= LENGTH(arrival)-1`
		Allow hard coded values for the number of items in the list.
		For example,
		`index >= 7`
		`index > 6`
			1

6(b)(ii)	Award **one** mark for any of the following: • `IF arrival[index] > 0` • `IF arrival[index] >= 1`		1
6(b)(iii)	Award **one** mark for: • `late` is incremented (1) `SET late TO late + 1`		1

Type MS · Spot selection and iteration, and pick the right loop

Answer 3 of 4 · Question 6 part a · 1 mark

2024 JUN · Paper 01 · extract 2 of 5 · find part a in this mark scheme extract

6(b)(iv)	Award **one** mark for:
	
	• On time is calculated as the remainder when early and late trains are subtracted from the total number of trains (1)
	`LENGTH(arrival) - (late + early)`
		Allow number of trains to be hard coded as 7 in place of
		`LENGTH(arrival)`
		
		Allow equivalent expressions:
		`LENGTH(arrival) - late - early`
			1

6(c)	Award **one** mark for any of the following: • Otherwise there would be an infinite loop (1) • It is used to index all elements in the (`arrival`) array (1) • To access the next value in the (`arrival`) array on the next iteration (1) • So that the condition used in the while loop evaluates to True when all the elements in arrival have been processed (1)		1

Type MS · Spot selection and iteration, and pick the right loop

Answer 3 of 4 · Question 6 part a · 1 mark

2024 JUN · Paper 01 · extract 3 of 5 · find part a in this mark scheme extract

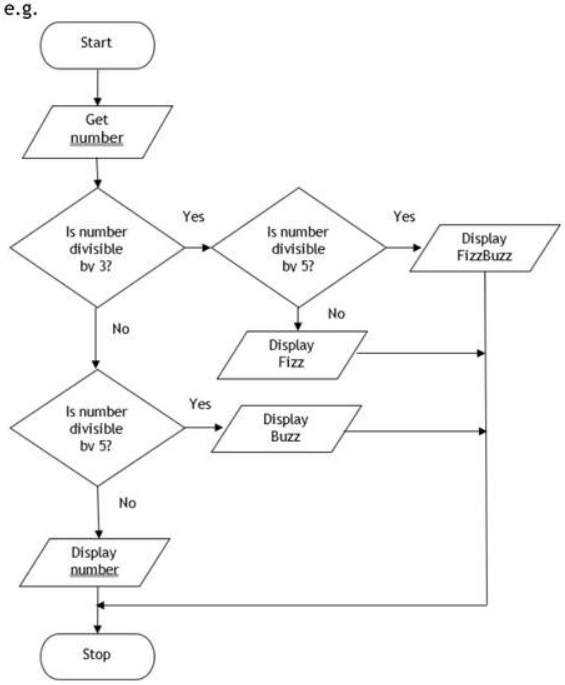
6(d)	Award **one** mark for each up to a maximum **six** marks: • Get number immediately follows start (1) • Number divisible by 3 and number divisible by 5 → Display Fizz (1) • Number divisible by 3 and number divisible by 5 → Display Buzz (1) • Number divisible by 3 and number divisible by 5 → Display FizzBuzz (1) • Number not divisible by 3 and number not divisible by 5 → Display n (1) • The flowchart represents a fully functional solution (1)	For last mark point: All symbols fully connected with directed arrows. Start and stop terminators used correctly. No hanging symbols. Decision boxes have exactly two labelled outputs For each of the first five mark points the mark should only be awarded if there is one correct logical path to get to the outcome. DO NOT credit a point if additional symbols have been added.	

Type MS · Spot selection and iteration, and pick the right loop

Answer 3 of 4 · Question 6 part a · 1 mark

2024 JUN · Paper 01 · extract 4 of 5 · find part a in this mark scheme extract

e.g.



			6
		Total for question 6	**12**

Type MS · Spot selection and iteration, and pick the right loop

Answer 3 of 4 · Question 6 part a · 1 mark

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Type MS · Spot selection and iteration, and pick the right loop

Answer 4 of 4 · Question 1 part a · 1 mark

2025 JUN · Paper 02 · extract 1 of 5 · find part a in this mark scheme extract

1(a)	A1	**The only correct answer is B** ***A*** *is not correct because for is used in iteration* ***C*** *is not correct because new is used to declare an object* ***D*** *is not correct because while is used in repetition*		**1**

Question	mp	Answer	Additional Guidance	Mark
1(b)(i)	B1 B2 B3	Award one mark for each correct cell:		**3**

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.		✔	
The program does not translate.	✔		
The program crashes during execution.			✔

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Type MS · Spot selection and iteration, and pick the right loop

Answer 4 of 4 · Question 1 part a · 1 mark

2025 JUN · Paper 02 · extract 2 of 5 · find part a in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(b)(ii)	B4	Line 15 - : added to end of line		
	B5	Line 20 – 0 changed to 1		
	B6	Line 22 – missing comma added between string and greenCount		3

```
# Q01bii

# initialise counters
ballCount = 0
greenCount = 0
redCount = 0

ballArray = ["green", "red", "green", "green", "red", "green","green", "green",
             "green", "green", "green", "red", "green", "red", "green", "red",
             "red", "green", "green", "red", "red", "green", "green", "red",
             "green", "red", "red", "green", "green", "red", "green", "green",
             "red", "green", "green", "green", "red", "green", "red","green"]

# iterate over ball array
for ball in ballArray:
    ballCount = ballCount + 1
    if ball == "green":
        greenCount = greenCount +1
    else:
        redCount = redCount + 1

print(" Green: ", greenCount, "balls \n", "Red: ", redCount,
      "balls \n", "Total: ", ballCount, "balls")
```

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Type MS · Spot selection and iteration, and pick the right loop

Answer 4 of 4 · Question 1 part a · 1 mark

2025 JUN · Paper 02 · extract 3 of 5 · find part a in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(c)(i)	C1	Award one mark for: • To store a value that must not be changed/prevent accidental change of value • To simplify code maintenance/so that if the value must be updated the change only has to made in a single place • To avoid inconsistency error of specifying same value multiple times • To make code more readable/give meaningful names to values	Accidental change cannot refer to 'user'	1

Question	mp	Answer	Additional guidance	Mark
1(c)(ii)	C2 C3	Award **two** marks for a linked description such as: • A variable stores a single value (1) whereas an array stores multiple values (1)		2

Question	mp	Answer	Additional guidance	Mark
1(d)(i)	D1	Award **one** mark for: • getWages (1)	Accept 'main' Accept: def getWages def getWages (inSales, inAge) line 7 (Python)	1

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Type MS · Spot selection and iteration, and pick the right loop

Answer 4 of 4 · Question 1 part a · 1 mark

2025 JUN · Paper 02 · extract 4 of 5 · find part a in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
1(d)(ii)	D2	Award **one** mark for: • and (1)	Do not accept line numbers Accept: &&	1

Question	mp	Answer	Additional guidance	Mark
1(d)(iii)	D3	Award **one** mark for any of the following: • sales (1) • wages (1) • inSales (1)	Do not accept line numbers	1

Question	mp	Answer	Additional guidance	Mark
1(d)(iv)	D4	Award **one** mark for any of the following: • inSales (1) • inAge (1)	Do not accept line numbers	1

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Type MS · Spot selection and iteration, and pick the right loop

Answer 4 of 4 · Question 1 part a · 1 mark

2025 JUN · Paper 02 · extract 5 of 5 · find part a in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark